

D E SHAW INTERNSHIP INTERVIEW EXPERIENCE

About the company- D. E. Shaw is a multinational investment management firm founded in 1988 by David E. Shaw and based in New York City. The company is known for developing complicated mathematical models and computer programs to exploit anomalies in financial markets.

Role Offered- SDET Intern (For the Quality and Test Engineering team based in Hyderabad)

Stipend- 1,50,000 INR per month

Selection process-

1. Online Assessment
2. Technical Interview 1
3. HR Interview
4. Technical Interview 2

Online Assessment

The Online Assessment was conducted on Hackerrank with a total duration of 95 minutes. The online assessment had 3 sections-

- Coding Questions (1st Question - 20 minutes & 2nd Question - 30 minutes)
- Technical MCQs (12 Questions - 17 minutes)
- Aptitude MCQs (14 Questions - 18 minutes)

Since each section had a separate time limit, it was necessary to complete the test sequentially and the extra time would not be carried forward for the remaining sections. The MCQs also had negative marks for incorrect answers.

Coding Question 1

Given an array of integers of size n , consider subarrays of a given size k . For every possible subarray of size k within the given array, calculate the sum of the absolute differences of all pairs of elements in that subarray and return the maximum sum in that array.

Example input –

Array - [1,2,3,5]

k - 3

Output-

8

I solved this question with a Brute Force approach after finding all possible subarrays of size k and finding sum of absolute differences for each subarray to find maximum sum of absolute differences. However, I was able to pass all example testcases without exceeding the time limit.

Coding Question 2

Given a string s and two empty strings pre and post, find the lexicographically smallest post string that can be formed by using all characters of string s by performing either of the two operations – removing an element from the beginning of s and adding it to the end of pre or removing an element from the end of pre and adding it to post.

Example input –

s – 'cab'

Output –

'abc'

I used a stack approach for moving characters in and out of the pre string and I used the Greedy approach to ensure that the lexicographically smallest string is the result. I was able to pass all example testcases with my approach.

Technical MCQs

The Technical MCQs were simple questions mostly based on DSA, DBMS, CN, finding errors in programs, predicting the output of a program and completing a portion of a given code.

Aptitude MCQs

The Aptitude related MCQs were a bit complex and included complex math questions too with the time limit making it harder to answer.

Based on the results of the Online Assessment, 17 people were shortlisted for the Interview.

Technical Interview Round 1 (1 hour)

The interview took place at CUIC. The interview started off with the panel having a look at my resume. Since I had completed a few projects related to DBMS and Data Science, they asked questions based on DBMS and to test my aptitude, problem solving skills.

- 1) If we were to create a database system for the Indian voting system, what tables should be included and with what form of normalization?

I was asked to write the relational schema of the tables on paper. After I gave them my answer, they asked me to consider adding an element of security to the database system. I told them about having an admin to control access. However, they were not satisfied with my answer and asked for a more secure database system. We moved on as I didn't know the answer.

- 2) What is the need for normalization and what are the different forms of normalization?

I told them the answer and gave them an example of when normalization would be required. They also questioned me about the normalization of the database I used in one of my projects.

- 3) You are tasked with using a water dispenser that can dispense cold, warm, and hot water to fill cups. Each type of water (cold, warm, hot) corresponds to an array position (amount[0], amount[1], and amount[2] respectively) indicating how many cups of each type you need to fill. The machine has two modes of operation-

- In one second, it can fill two cups of different types of water.
- In one second, it can fill one cup of any type of water.

The goal is to find the minimum number of seconds required to fill all the cups as per the amount array.

I explained my approach of filling two maximum types at a point of time and moving on to the next type after finishing off with atleast one type. They were satisfied with my approach and I was asked to write the code on paper.

4) Given an increase in the number of IPL teams, it was decided to divide the total number of teams in two groups and the matches played by each team would be based on the following conditions-

- 1 match with every other team in the same group
- 2 matches with other teams in the other group

Given n number of teams, find the total number of matches conducted.

It was more of a mathematical aptitude question as they asked me to find the answer for 10 teams. I gave the wrong answer at first but I was able to correct myself later. I was asked to find a general formula for it considering n number of teams and write the code for a function to implement the same on paper.

5) Given a table with Employee name, ID, Job name, Job ID, State Code, State with several repetitive values in all the attributes, I was asked to find a suitable primary key, possible functional dependencies and the normal form. I was also asked to normalize the table and write relational schemas of the final normalized forms on paper.

We also had general discussions about the internship during the interview. At the end of the interview, they asked me if I had any questions for them. I asked them for any suggestions to improve my skills. They advised me to look for more creative and unexpected solutions for coding questions instead of the usual solutions.

HR Interview (15 minutes)

The interview was more of a friendly discussion with the HR. I was asked questions like-

- 1) Why do you prefer an internship related to testing instead of developing?
- 2) What made you choose D. E. Shaw for an internship?
- 3) If given a better opportunity in another company, would you prefer to stay or leave?
- 4) What are your plans for higher studies?
- 5) What do you know about D. E. Shaw and what did you get to know from the Pre Placement Talk?
- 6) Would you prefer to work in a new environment far away from your friends and family?

I told the HR about why I was interested in a FinTech related internship. I explained how I believed testing and developing would require equal levels of creativity and so it doesn't matter whether it's testing or developing. I also told her that a Software related internship in a company would be a dream come true and factors like a new work environment wouldn't affect me much. I also said that better opportunities, higher studies are all a part of a future plan but I would like to gain some hands-on experience before making any important life decisions.

After the first technical interview and the HR interview, few of us were asked to attend the second technical interview in the evening.

Technical Interview Round 2 (45 minutes)

The interview started off with the panel asking me to introduce myself. After the introduction, they moved on to the DSA questions.

- 1) Given a list of library books and a list of students with each student having read a few books, find the minimal set of students that must have read maximum/almost all the library books together.

I gave the wrong answer at first as I misunderstood the question. However, the interviewer was able to correct me and I was able to give the right answer eventually. I explained my logic of using the sliding window approach considering the student roll numbers were in order and books read by each student was stored in a dictionary with their roll numbers as keys. I also explained another approach using Kadane's algorithm as the interviewer was looking for an optimized approach. I was questioned about the time and space complexity of my approach and asked for a more optimized answer. We moved on as I couldn't think of a more optimized answer.

- 2) There are 3 repositories in a github account. Each repository has around 50 files and testing a file would require an enormous amount of time. Considering the problem that raises an error involving a combination of one file in each of the repository due to a bug in one of the files in the 3 repositories, suggest an approach to find the file with the bug without taking much time.
I explained my approach of using binary search considering the fact that a bug in a particular file will also affect the working of the subsequent files. I was asked about the time complexity of my approach and they were satisfied with my approach.

- 3) Given a $m \times n$ matrix with integers, find a path providing the maximum sum from the top left corner to the bottom right corner by moving diagonally up or diagonally down or towards the right and without returning to the same spot more than once.

I gave them a backtracking approach considering movements in the diagonally up, diagonally down and right directions. I was asked about the time and space complexity. I had to write the code on paper and they were satisfied with my approach.

- 4) The question was about ball collision similar to the asteroid collision available on leetcode.

<https://leetcode.com/problems/asteroid-collision/description/>

I explained a stack approach for it and wrote the code on paper. They were satisfied with my answer after I explained the time and space complexities.

After the DSA questions, they had a look at my resume and inquired generally about my achievements. As I had mentioned C++ and Python as my skills, they asked me about the difference between the two languages. I explained about the differences of the compiled and interpreted language. They also asked me which language would be better in different situations involving programs with time as an important factor, programs that need not be precompiled again and again and so on.

At the end of the interview, they asked me about why I chose to dive deep into a few specific languages instead of exploring more. They also advised me to improve myself with newer technological skills.

The results were announced later at night and 2 of us from CEG were selected for the internship.

General Tips-

It's completely fine if you are not a great coder in the start. You always have time to improve your skills.

- Practice coding in websites like Leetcode, Hackerrank as it helps improve problem solving skills.
- Have a look at questions in Leetcode Premium for the company.
- Practice ACM CODE questions for an understanding of every single topic in DSA.

- Learn from the experiences of seniors.
- Try practicing aptitude questions also as it has great importance in the online assessment.
- Improve your speaking skills. The interview will be more of a discussion rather than an oral test and the interviewer also notices your ability to interact easily.
- The interviewer is more interested to know your creative answers to their coding questions. So let your creativity flow while finding optimized solutions.
- Keep revising old concepts while learning new ones.

Last but not the least.... Stay calm!

All the best!

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